

CANONICAL LESSON

Rainfall and Drought Impact Monitoring

Decision-centred foundations for accountable sector intelligence

Production Unit	PU-B07-C13
Notebook	V7_A_N08
Status	Controlled educational draft v0.1
Data boundary	Synthetic instructional data only

Central decision: Link rainfall anomalies to exposure and observed agricultural impacts before escalating drought-related decisions.

Learning outcomes

Why this matters

Rainfall deficit alone does not prove crop loss, household impact, or the need for a specific intervention. The analytical workflow is useful only when evidence, uncertainty, authority, and action are connected explicitly. The model supports professional judgement; it does not replace accountable institutional authority.

Worked cross-sector example

A rainfall deficit is combined with crop calendars, geographic coverage, and field observations before a district is referred for review.

Boundary: The example supports learning and review. It does not constitute an official statistic, eligibility determination, enforcement action, clinical judgement, or operational authorization.

Practice and reflection

- 1.
- Add a standardized precipitation index with a documented reference period.
- 2.
- Compare station-only and satellite-assisted results.
- 3.
- Add crop-specific calendars.

- 4.
- Explain why a drought alert cannot determine compensation.

Answer key

- 1.
- Fit/standardize precipitation over an approved long reference series and report distribution assumptions.
- 2.
- Evaluate bias, coverage, spatial representativeness, and disagreement; route large disagreement to verification.
- 3.
- Join locally validated crop-stage dates and propagate calendar uncertainty.
- 4.
- Compensation also requires legal criteria, verified losses, exposure, eligibility, budgets, appeals, and authorized decisions.

Operational readiness checklist

- Purpose and decision owner are named.
- Evidence definitions, provenance, and lawful access are documented.
- Uncertainty and abstention are visible.
- Human review, escalation, monitoring, incident response, and retirement are assigned.
- Equity, accessibility, security, and sector-specific approval are complete.